Practice Set B

Class 12 Mathematics, Chapter 13: Probability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, NCERT Exemplar — 1 mark

If $P(B) = \frac{3}{5}$, $P(A|B) = \frac{1}{2}$ and $P(A \cup B) = \frac{4}{5}$, then $P(A \cup B)' + P(A' \cup B)$ is equal to:

(A) $\frac{1}{5}$ **(B)** $\frac{4}{5}$ **(C)** $\frac{1}{2}$ **(D)** 1

2 MCQ, NCERT Exemplar — 1 mark

Three persons A, B and C fire at a target in turn, starting with A. Their probabilities of hitting the target are 0.4, 0.3 and 0.2 respectively. The probability of two hits is:

(A) 0.024 **(B)** 0.188 **(C)** 0.336 **(D)** 0.452

3 MCQ, NCERT Exemplar — 1 mark

Two dice are thrown. If it is known that the sum of numbers on the dice was less than 6, the probability of getting a sum 3 is:

(A) $\frac{1}{18}$ **(B)** $\frac{5}{18}$ **(C)** $\frac{1}{5}$ **(D)** $\frac{2}{5}$

4 MCQ, NCERT Exemplar — 1 mark

If A and B are such events that $P(A) > 0$ and $P(B) \neq 1$, then $P(A'|B')$ equals:

(A) $1 - P(A|B)$ **(B)** $1 - P(A'|B)$

(C) $\frac{1 - P(A \cup B)}{P(B')}$ **(D)** $\frac{P(A')}{P(B')}$

5 MCQ, NCERT Exemplar — 1 mark

Two events E and F are independent. If $P(E) = 0.3$ and $P(E \cup F) = 0.5$, then $P(E|F) - P(F|E)$ equals:

(A) $\frac{2}{7}$ **(B)** $\frac{3}{35}$ **(C)** $\frac{1}{70}$ **(D)** $\frac{1}{7}$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** If $P(A) = 0.3$ and $P(B) = 0.4$, then A and B cannot be both mutually exclusive and independent. **Reason (R):** If A and B are mutually exclusive, then $P(A \cap B) = 0$; if they are independent, then $P(A \cap B) = P(A)\,P(B)$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Exemplar A bag contains 5 red marbles and 3 black marbles. Three marbles are drawn one by one without replacement. What is the probability that at least one of the three marbles drawn be black, if the first marble is red?	2 marks

Level 2

18 marks

8	Short answer, NCERT Ex 13.3 Q7 An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probabilities of an accident are 0.01, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver?	3 marks
9	Short answer, NCERT Exemplar A bag contains 4 white and 5 black balls. Another bag contains 9 white and 7 black balls. A ball is transferred from the first bag to the second and then a ball is drawn at random from the second bag. Find the probability that the ball drawn is white.	2 marks
10	Short answer, Board-pattern For two events A and B, it is given that $P(A) = 0.6$, $P(A \cup B) = 0.8$ and $P(A\|B) = 0.5$. Find $P(B)$, $P(B\|A')$ and $P(A'\|B')$.	3 marks
11	Short answer, CBSE SQP 2022-23 Three friends go for coffee. They decide who will pay the bill, by each tossing a coin and then letting the “odd person” pay. There is no odd person if all three tosses produce the same result. If there is no odd person in the first round, they make a second round of tosses and they continue to do so until there is an odd person. What is the probability that exactly three rounds of tosses are made?	3 marks
12	Short answer, NCERT Exemplar Assume that in a family, each child is equally likely to be a boy or a girl. A family with three children is chosen at random. Find the probability that the eldest child is a girl, given that the family has at least one girl.	2 marks

Q No.	Question	Marks
13	Short answer, NCERT Exemplar A bag contains $(2n+1)$ coins. It is known that n of these coins have a head on both sides, whereas the rest of the coins are fair. A coin is picked up at random from the bag and is tossed. If the probability that the toss results in a head is $\frac{31}{42}$, determine the value of n.	3 marks
14	Short answer, NCERT Exemplar Two dice are tossed. Find whether the following two events A and B are independent: $A=\{(x,y):x+y=11\}$, $B=\{(x,y):x\neq 5\}$, where (x,y) denotes a typical sample point.	2 marks

Level 3 — 25 marks

Q No.	Question	Marks
15	Long answer, NCERT Exemplar A shopkeeper sells three types of flower seeds A_1, A_2 and A_3. They are sold as a mixture where the proportions are $4:4:2$ respectively. The germination rates of the three types of seeds are 45%, 60% and 35%. Calculate the probability **(i)** of a randomly chosen seed to germinate, — 2 marks **(ii)** that it will not germinate given that the seed is of type A_3, — 1 mark **(iii)** that it is of the type A_2 given that a randomly chosen seed does not germinate. — 2 marks	5 marks
16	Short answer, PYQ-style In a test, an examinee either guesses or copies or knows the answer to a multiple-choice question with four choices. The probability that he makes a guess is $\frac{1}{3}$ and the probability that he copies the answer is $\frac{1}{6}$. The probability that his answer is correct, given that he copied it, is $\frac{1}{8}$. Find the probability that he knew the answer to the question, given that he correctly answered it.	3 marks
17	Long answer, NCERT Ex 13.3 Q10 Suppose a girl throws a die. If she gets a 5 or 6, she tosses a coin three times and notes the number of heads. If she gets 1, 2, 3 or 4, she tosses a coin once and notes whether a head or tail is obtained. If she obtained exactly one head, what is the probability that she threw 1, 2, 3 or 4 with the die?	5 marks
18	Short answer, NCERT Exemplar Three events A, B and C have probabilities $\frac{2}{5}$, $\frac{1}{3}$ and $\frac{1}{2}$, respectively. Given that $P(A\cap C)=\frac{1}{5}$ and $P(B\cap C)=\frac{1}{4}$, find the values of $P(C\|B)$ and $P(A'\cap C')$.	3 marks
19	Long answer, NCERT Ex 13.3 Q12 A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and are found to be both diamonds. Find the probability of the lost card being a diamond.	5 marks

Q No.	Question	Marks
20	Case study, CBSE SQP 2024-25 **Case Study:** Arka bought two cages of birds: Cage-I contains 5 parrots and 1 owl and Cage-II contains 6 parrots. One day Arka forgot to lock both cages and two birds flew from Cage-I to Cage-II (simultaneously). Then two birds flew back from Cage-II to Cage-I (simultaneously). Assume that all the birds have equal chances of flying. On the basis of the above information, answer the following questions. **(i)** When two birds flew from Cage-I to Cage-II and two birds flew back from Cage-II to Cage-I, find the probability that the owl is still in Cage-I. — 2 marks **(ii)** When two birds flew from Cage-I to Cage-II and two birds flew back from Cage-II to Cage-I, the owl is still seen in Cage-I. What is the probability that one parrot and the owl flew from Cage-I to Cage-II? — 2 marks	4 marks

Answer key

Chapter 13: Probability, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) 1
2	(B) 0.188
3	(C) $\frac{1}{5}$
4	(C) $\frac{1-P(A\cup B)}{P(B')}$
5	(C) $\frac{1}{70}$
6	(A) Both A and R are true and R is the correct explanation of A
7	$\frac{5}{7}$

Level 2

8	$P(E_1\|A)=\frac{1}{52}$
9	$\frac{5}{9}$
10	$P(B)=0.4$, $P(B\|A')=0.5$, $P(A'\|B')=\frac{1}{3}$
11	$\frac{3}{64}$
12	$\frac{4}{7}$
13	$n=10$
14	Not independent, since $P(A\cap B)=\frac{1}{36}\neq\frac{5}{108}=P(A)\,P(B)$

Level 3

15	(i) 0.49 (ii) 0.65 (iii) $\frac{16}{51}$
16	$P(E_1\|A)=\frac{24}{29}$
17	$P(E_2\|A)=\frac{8}{11}$
18	$P(C\|B)=\frac{3}{4}$, $P(A'\cap C')=\frac{3}{10}$
19	$P(E_1\|A)=\frac{11}{50}$
20	(i) $\frac{3}{4}$ (ii) $\frac{1}{9}$